Lund University

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Course name: Image Analysis

Assignment 2

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1 Filtering

Before starting, two remarks.

The first is about the convention we use for convolution. With a 3×3 filter we place the centre coefficient on top of the pixel we are computing, multiply each coefficient by the pixel underneath it and sum the products. Two of the filters here have no centre coefficient, since f_1 is 3×4 and f_5 is 1×10 and an even-sized axis has no middle element. What the filter actually needs is not a centre but an *origin*: one coefficient chosen to sit on the output pixel, with the others indexed as offsets from it, so that

$$v(r, c) = \sum_{i,j} f(i, j)\, u(r - i,\, c - j).$$

For an odd size the natural origin is the centre, which gives back the usual rule. For an even size we have to pick one of the two middle coefficients, and the two choices give the same result translated by one pixel. We use the convention of `scipy.signal.convolve2d` with `mode='same'`, which places the origin at coefficient $\lfloor (K-1)/2 \rfloor$ along a K-long axis. The choice does not affect any of the conclusions below, because a one-pixel translation cannot change whether a result is a blur, an inversion or an edge image; it only fixes in which direction image D is displaced and from which diagonal image E appears to be lit.

The second remark is about why two of the results are uniformly grey. If the coefficients sum to zero, every region where the image is locally constant maps to exactly zero, since the positive and negative contributions cancel. The results are displayed with automatic grey-scale scaling, so the smallest value is drawn black and the largest white, and zero therefore lands in the middle of the scale. A flat grey area consequently does not mean that the output was grey, it means that the output was *zero*, i.e. that the filter responded to nothing there.

> Before making the final matching, classify the filters qualitatively (for example, smoothing, sharpening, edge/difference detection, or spatial shift) and state what visual effect you expect from each one.

Three properties of a filter are enough to predict its behaviour: the sum of the coefficients, their signs, and the symmetry of the matrix.

The sum determines the gain on constant regions. If the image takes the same value c everywhere inside the filter window, the output is c times the sum. A sum of 1 therefore preserves brightness, a sum of 0 maps every constant region to zero so that only variation can produce a response, and a negative sum maps bright to dark, which inverts the image.

The signs determine whether the filter averages or differentiates. If all coefficients share a sign, the operation is a weighted average of the neighbourhood, and averaging attenuates high frequencies, so the image is smoothed. If both signs occur, the filter subtracts one part of the neighbourhood from another, so it responds to variation rather than to absolute intensity.

The symmetry determines what a feature looks like in the result. Rotating a filter by 180° pairs every coefficient with the one at the opposite offset, and what matters is the relative sign within each pair. If the two have opposite signs the filter is antisymmetric, so it subtracts one side of the neighbourhood from the other, which is a directional derivative: the response cancels wherever the image is locally symmetric and takes opposite signs on the two sides of a feature, leaving one flank brighter than the background and the other darker, which we read as relief. If the two have the same sign the filter is symmetric and adds the sides instead, so nothing cancels and both flanks are rendered identically, which gives a plain outline.

Applying this to the six filters: f_2 has only non-negative coefficients summing to 1, so it is a smoothing filter that preserves brightness, and we expect a blurred version of the original. Filter f_6 has only negative coefficients summing to -1, so it is the same kind of average with the sign reversed, and we expect a slightly blurred negative. Filter f_4 also sums to 1 but combines

a large positive centre coefficient with a negative surround, which is a sharpening filter, so we expect something almost indistinguishable from the original apart from crisper edges. Filter f_5 contains a single coefficient equal to 1, so it is a shifted unit impulse and each output pixel is a copy of one input pixel; we expect a pure translation. Finally f_1 and f_3 both sum to 0, so both give a mid-grey field with only edges drawn on it. They differ in symmetry: f_1 is symmetric, so we expect plain outlines, whereas f_3 is antisymmetric, so we expect the embossed appearance of a surface lit from one direction.

Match each output image to the corresponding filter and motivate each match from the structure of the filter, not only from visual similarity.

There are six filters and only five images, so one filter is unused.

Image A is inverted, since the dark hair has become white and the bright background dark. Inversion is decided by the coefficient sum, and f_6 is the only filter of the six whose sum is negative, so A is f_6. Image B is a blurred version of the original at unchanged overall brightness, which is exactly a non-negative weighted average with sum 1, so B is f_2. Image D has the same sharpness, contrast and brightness as the original and differs only by a displacement, and the only filter that leaves all of those properties untouched is the one that merely copies single pixels, so D is f_5. Images C and E are the two zero-sum results and must therefore be f_1 and f_3 in some order. In C only vertical structure survives and the horizontal strokes of the letters in "ADELE 21" have vanished, which matches f_1; E is embossed, which matches f_3.

The filter left over is f_4, and this is consistent, since f_4 sharpens and its output would look almost identical to the original image. No such image appears among A–E: B is blurred and D is displaced.

For at least two of the filters, explain how the signs, symmetry, and sum of the filter coefficients support your interpretation.

For $f_6 = -\frac{1}{9}\mathbf{1}_{3\times 3}$ all nine coefficients are equal and negative. Because the magnitudes are equal, the filter is the ordinary 3×3 box average multiplied by -1, so it smooths and inverts at the same time. The sum is -1, which means a constant region of intensity c is mapped to $-c$. This is why A is a negative, and since no other filter has a negative sum, no other filter could have produced it.

For f_1 the three rows are identical, so the matrix has rank one and is separable:

$$f_1 = \frac{1}{6}\begin{bmatrix}1\\1\\1\end{bmatrix}\begin{bmatrix}1 & -1 & -1 & 1\end{bmatrix}.$$

This is the key structural fact, because it decouples the two directions. In the y direction the filter is $[1, 1, 1]$, a plain average, and an average cannot detect variation, it can only smooth it. The filter is therefore blind to horizontal edges. All of its selectivity comes from the row $[1, -1, -1, 1]$, which acts along x only, so only vertical structure survives. The sum is 0, which accounts for the grey background, and the row is symmetric, so it behaves as a second derivative and renders each line as a double outline rather than giving it a bright and a dark side.

For f_3 the sum is also 0, so the background is grey as well, but the symmetry is the opposite: rotating the matrix by 180° gives its negative. The positive coefficients lie on the upper-right side of the main diagonal and the negative ones on the lower-left, so the filter is a directional first derivative along the direction from the lower-left corner towards the upper-right corner. A first derivative is signed, so an edge facing one way is rendered brighter than the grey background and an edge facing the other way darker. Light on one side of a feature and dark on the other is interpreted by the visual system as a surface under directional illumination, which is why E appears embossed.

Identify one match that is especially easy to distinguish and one that is potentially more ambiguous, and explain why.

The easiest match is A with f_6, and it can be settled from a single number. The coefficient sum is the gain applied to constant regions, and f_6 is the only filter whose sum is negative, so it is the only one capable of inverting intensities. Image A is unambiguously inverted, so the match follows without examining any detail.

The difficult pair is C and E, that is f_1 against f_3. They agree on the properties that govern the overall appearance: both sum to zero, so both produce a mid-grey field carrying only edge information, and both have small support, so both produce a thin and faint response. At the printed size the two images are the same kind of picture, and neither the sum nor the sign pattern distinguishes them.

What does distinguish them is orientation selectivity and symmetry. Because f_1 contains an average in the y direction, its response to a purely horizontal edge is not merely weak but identically zero.

2 Interpolation

The signal is $f = \begin{bmatrix} 1 & 2 & 2 & 4 & 7 & 6 & 5 \end{bmatrix}$, sampled at the points $i = 1, \ldots, 7$.

Explain what linear interpolation means. Sketch the linear interpolation $F_{lin}(x)$ of f for $1 \leq x \leq 7$. Is $F_{lin}(x)$ continuous? Is $F_{lin}(x)$ differentiable?

Linear interpolation means joining neighbouring samples by straight line segments. For x in $[i, i+1]$, writing $t = x - i$ so that t runs from 0 to 1 across the interval,

$$F_{lin}(x) = (1 - t)\, f(i) + t\, f(i+1),$$

a weighted average of the two surrounding samples in which each weight is the closeness to that sample.

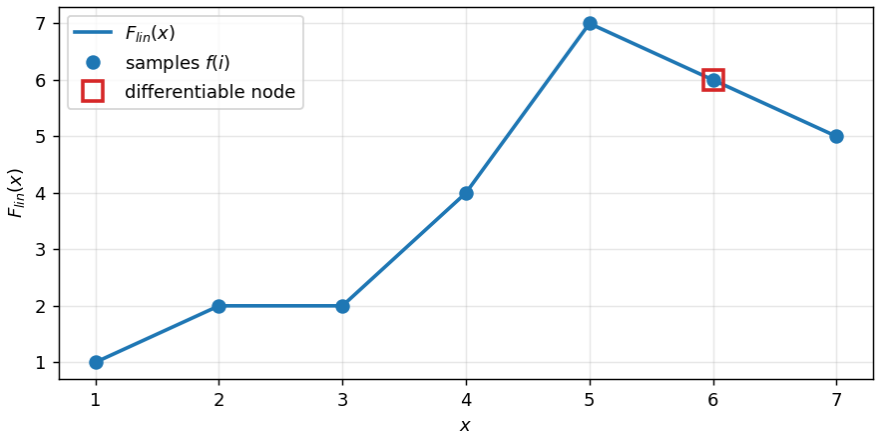


Figure 1: The linear interpolation of f. The square marks $x = 6$, the one interior node where the interpolant is differentiable.

It is continuous. Each piece is a polynomial, so continuity can only fail at a node, and at $x = i$ the piece on the left ends at $f(i)$ while the piece on the right starts at $f(i)$, so the two one-sided limits agree.

It is not differentiable everywhere. Inside an interval the interpolant is a straight line, but at a node the one-sided derivatives are $F'_{lin}(i^-) = f(i) - f(i-1)$ and $F'_{lin}(i^+) = f(i+1) - f(i)$, which agree only when the data has the same difference on both sides. Since the samples are one apart, each difference $f(i+1) - f(i)$ is exactly the slope of that segment, and the six slopes are $+1, 0, +2, +3, -1, -1$. They disagree at $x = 2, 3, 4, 5$ and agree at $x = 6$, where both are -1. Differentiability therefore fails only at those nodes where the slope actually changes, which depends on the data rather than on the method.

Which function $g(x)$ corresponds to linear interpolation? Give the function $g(x)$ for $-4 \leq x \leq 4$.

It is the triangle, or hat, function

$$g(x) = \begin{cases} 1 - |x| & \text{if } |x| < 1, \\ 0 & \text{otherwise,} \end{cases}$$

so on $-4 \leq x \leq 4$ it is the triangle on $(-1, 1)$ and zero elsewhere.

To check it, take x in $[i, i+1]$ and put $t = x - i$. Every term of $F_g(x) = \sum_k g(x-k)f(k)$ vanishes except $k = i$ and $k = i+1$, since all other arguments exceed 1 in absolute value. The two surviving weights are $g(t) = 1 - t$ and $g(t-1) = t$, giving $(1-t)f(i) + tf(i+1)$, which is F_{lin}. Two properties matter for the comparison later: $g(0) = 1$ with $g(k) = 0$ at every other integer, which makes the curve pass through the data, and support $[-1, 1]$, so only two samples affect any value.

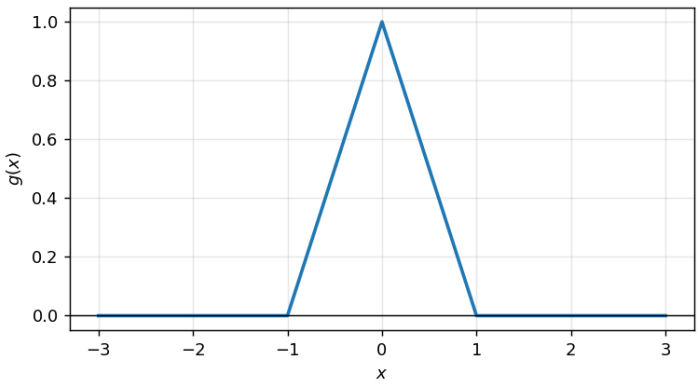


Figure 2: The triangle kernel, the g that reproduces linear interpolation.

Let $g(x) = \frac{6}{7}(1 - |x|^2)$ for $|x| \leq 1$, $g(x) = -\frac{6}{7}(|x|^3 - 5|x|^2 + 8|x| - 4)$ for $1 < |x| \leq 2$, and $g(x) = 0$ for $|x| > 2$. Plot $g(x)$ for $-3 \leq x \leq 3$. Plot the interpolation F_g of f using this g, for $1 \leq x \leq 7$. Is $F_g(x)$ continuous? Is $F_g(x)$ differentiable?

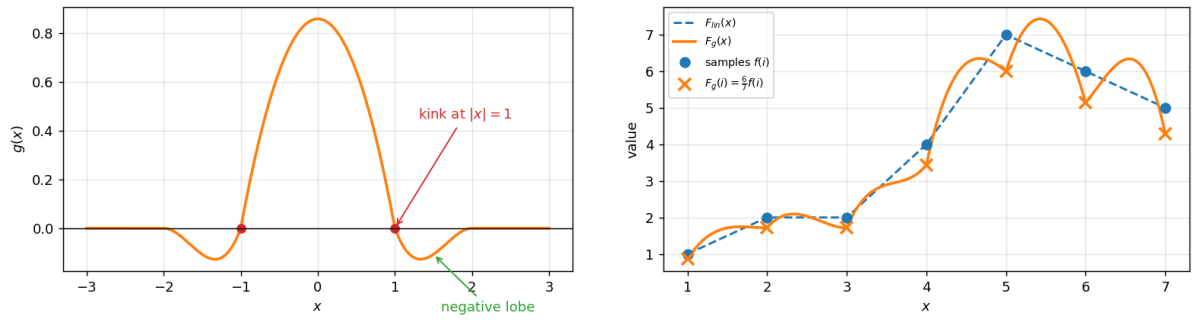


Figure 3: Left: the kernel g, with the kink at $|x| = 1$ and the negative side lobes marked. Right: the interpolation F_g (solid) with the linear interpolation (dashed) and the samples; the crosses show that F_g passes not through $f(i)$ but through $\frac{6}{7}f(i)$.

Since the answers depend on the behaviour of g at its joining points, we inspect those first. Writing $s = |x|$, the two joins are at $s = 1$ and $s = 2$, and both pieces vanish at each of them, so g is continuous. Differentiating the pieces as given, $\frac{6}{7}(1 - s^2)$ has derivative $-\frac{12s}{7}$ and $-\frac{6}{7}(s^3 - 5s^2 + 8s - 4)$ has derivative $-\frac{6}{7}(3s^2 - 10s + 8)$, so

join	from the left	from the right	
$s = 1$	$-12/7$	$-6/7$	kink
$s = 2$	0	0	differentiable

Table 1: One-sided derivatives of g at its joining points.

So g is continuous everywhere and differentiable everywhere except at $|x| = 1$, where g' jumps by $6/7$. (The join at $s = 2$ is smooth because the cubic factors as $(s - 1)(s - 2)^2$: the double root makes both g and g' vanish there, whereas the simple root at $s = 1$ makes only g vanish.)

These properties carry over to $F_g = \sum_{i=1}^{7} g(x - i)f(i)$, which is a finite sum of shifted, scaled copies of g. A finite sum of continuous functions is continuous, so F_g **is continuous everywhere**.

For differentiability, the term $g(x - i)$ can only be non-differentiable where its argument reaches ± 1, that is at $x = i \pm 1$, so all possible kinks of F_g sit at integers and everywhere else F_g is a finite sum of differentiable functions. At a node $x = m$ exactly two terms kink, the ones with $i = m - 1$ and $i = m + 1$, whose arguments pass through $+1$ and -1. The jump in g' is $6/7$ at both places, since g is even and therefore g' is odd, so

$$F_g'(m^+) - F_g'(m^-) = \tfrac{6}{7}\big[\, f(m - 1) + f(m + 1) \,\big].$$

At $x = 4$, for instance, only the terms $i = 3$ and $i = 5$ kink and the jump is $\frac{6}{7}(2{+}7) = 54/7 \approx 7.71$. All samples are positive, so the expression is never zero and F_g **is not differentiable at any interior node**. The jumps at $x = 2, \ldots, 6$ are $18/7, 36/7, 54/7, 60/7$ and $72/7$, and one-sided difference quotients computed numerically reproduce these to four decimals.

Finally, note that $g(0) = \frac{6}{7}$ rather than 1, while $g(\pm 1) = g(\pm 2) = 0$. At a sample point only one term survives, so $F_g(i) = \frac{6}{7}f(i)$ and the curve misses every data point by that factor, as the crosses in Figure 3 show. Strictly this g is a smoothing kernel rather than an interpolating one; the factor $6/7$ normalises $\int g = 1$ instead of $g(0) = 1$.

Compare the interpolation obtained with this g with the linear interpolation: comment on locality, smoothness, and how values between the samples are influenced by neighbouring samples.

Locality. The triangle kernel has support $[-1, 1]$, so $F_{lin}(x)$ depends on the two samples bracketing x. The kernel g has support $[-2, 2]$, so $F_g(x)$ generally depends on four: at $x = 3.5$ the samples $f(2), \ldots, f(5)$ all contribute, against only $f(3)$ and $f(4)$ for linear interpolation. Wider support means a local error in the data spreads further.

Smoothness. Between the nodes F_g is a cubic and therefore curved, so it looks smoother than the straight segments of F_{lin}, but at the nodes both are only continuous. F_{lin} kinks wherever the data changes slope, while F_g kinks at every node, including $x = 6$ where F_{lin} does not. By the strict criterion the cubic construction is thus the less regular of the two here: what it buys is curvature between the nodes, not differentiability at them.

Influence of neighbouring samples. The decisive property is the sign of the kernel. The triangle kernel is non-negative with weights summing to 1, so F_{lin} is a convex combination of two data values and can never leave the range of the data. The kernel g is negative on $1 < |x| < 2$, so the outer samples enter with negative weight and push the curve away from them. Two consequences are visible in Figure 3. The data is flat on $[2, 3]$, where $f(2) = f(3) = 2$, and F_{lin} is correspondingly exactly flat, but F_g bulges to 2.095 at $x \approx 2.33$ because $f(1)$ and $f(4)$ act through the negative lobes. And although the largest sample is $f(5) = 7$, F_g reaches 7.43 at $x \approx 5.43$, overshooting the range of the data, which non-negative weights cannot do. This overshoot is the one-dimensional version of the bright and dark fringes a sharpening filter produces along an edge.

3 Classification using Nearest Neighbour and Bayes theorem

The measurements are

Class 1	0.4003	0.3985	0.3998	0.3997	0.4015	0.3995	0.3991
Class 2	0.2554	0.3139	0.2627	0.3802	0.3247	0.3360	0.2974
Class 3	0.5632	0.7687	0.0524	0.7586	0.4443	0.5505	0.6469

Before computing anything it is worth looking at how the three groups sit on the axis, since that already predicts where the difficulty will be. Alongside the measurements we also have the Gaussian model of Section 3.2, which specifies the class means and standard deviations as $m_1 = 0.40$, $\sigma_1 = 0.01$; $m_2 = 0.32$, $\sigma_2 = 0.05$; and $m_3 = 0.55$, $\sigma_3 = 0.2$. These are given as known parameters rather than estimated from the data, and they are close to but not identical with what we would estimate from the measurements themselves. Using $\hat{\mu} = \frac{1}{N}\sum_i x_i$ and $\hat{\sigma}^2 = \frac{1}{N}\sum_i (x_i - \hat{\mu})^2$, the seven samples of each class give $\hat{\mu} = 0.3998,\ 0.3100,\ 0.5407$ and $\hat{\sigma} = 0.0009,\ 0.0400,\ 0.2265$.

Class 1 is extraordinarily concentrated: all seven values lie between 0.3985 and 0.4015, a span of 0.0030. Class 2 is moderately spread, from 0.2554 to 0.3802 about its mean of 0.32. Class 3 is spread very widely, from 0.0524 to 0.7687 with $\sigma_3 = 0.2$, so its samples straddle both of the other two classes. We therefore expect class 1 to be the easiest, since it occupies a tiny interval that no other class concentrates on, and class 3 to be the hardest, since some of its samples fall closer to the other two classes than to the bulk of their own. Class 2 should be intermediate, with its upper tail reaching towards class 1.

3.1 Nearest Neighbours

Use nearest neighbour classification. Assume that the first four measurements in each class are used for training and the last three for testing. How many measurements will be correctly classified?

With four training and three test samples per class there are twelve training points and nine test points. Each test point is given the label of the training point nearest to it. **Eight of the nine are correct.** All three test samples of class 1 and all three of class 2 are classified correctly; of class 3, two are correct and $x = 0.4443$ is not.

As a worked example, take that failing point. Its distance to the nearest training sample of each class is

	nearest training sample	distance
class 1	0.4003	**0.0440**
class 2	0.3802	0.0641
class 3	0.5632	0.1189

The smallest distance belongs to class 1, so the point is assigned to class 1 although it truly belongs to class 3. The error is easy to explain from the training data. The four training samples of class 3 are 0.0524, 0.5632, 0.7586 and 0.7687, which leaves a wide empty stretch between 0.05 and 0.56, and 0.4443 falls inside it. The four training samples of class 1, by contrast, all lie within 0.002 of 0.4, so any point in that neighbourhood is close to one of them. Nearest neighbour compares only raw distances to individual samples, so a sparsely sampled class with a gap loses to a densely sampled class nearby.

3.2 Gaussian distributions

Assume that the measurements have normal distribution with known parameters (expected value m_i and variance σ^2): $m_1 = 0.4$, $\sigma_1 = 0.01$; $m_2 = 0.32$, $\sigma_2 = 0.05$; $m_3 = 0.55$, $\sigma_3 = 0.2$. All classes are equally likely to occur. What is the maximum a posteriori classification of a measured x? How many measurements are correctly classified?

The rule is to pick the class with the largest posterior,

$$\hat{\jmath} = \arg\max_k P(Y = k \mid x),$$

and Bayes' theorem rewrites that posterior in terms of quantities we can actually compute,

$$P(Y = k \mid x) = \frac{f_X(x \mid Y = k)\, P(Y = k)}{f_X(x)}, \qquad f_X(x) = \sum_j f_X(x \mid Y = j)\, P(Y = j).$$

The denominator $f_X(x)$ does not depend on k, so it is one and the same number for all three classes and cannot affect which posterior is largest; only the numerators need to be compared. Here the classes are in addition equally likely, $P(Y = k) = 1/3$, so the prior is a common factor as well and also drops out. What remains is to choose the largest likelihood

$$f_X(x \mid Y = k) = \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(x - m_k)^2}{2\sigma_k^2} \right).$$

No measurements were used for training, so all twenty-one are classified. **Twenty of the twenty-one are correct**: classes 1 and 3 are classified perfectly and the single error is $x = 0.3802$, a class 2 sample assigned to class 1.

For that point the three likelihoods are

$$f_X(x \mid Y=1) = \frac{1}{0.01\sqrt{2\pi}}\, e^{-1.98^2/2} = 39.89 \times 0.1408 = 5.618,$$
$$f_X(x \mid Y=2) = \frac{1}{0.05\sqrt{2\pi}}\, e^{-1.204^2/2} = 7.98 \times 0.4844 = 3.865,$$
$$f_X(x \mid Y=3) = \frac{1}{0.20\sqrt{2\pi}}\, e^{-0.849^2/2} = 1.99 \times 0.6974 = 1.391,$$

so class 1 wins. This deserves a comment, because measured in units of each class's own standard deviation the point is 1.98σ away from class 1 but only 1.20σ away from class 2, so counting standard deviations alone would have favoured class 2. Taking logarithms,

$$\ln f_X(x \mid Y=k) = -\frac{(x-m_k)^2}{2\sigma_k^2} - \ln \sigma_k + \text{const},$$

which shows that the decision is not made on the number of standard deviations alone: the term $-\ln \sigma_k$ gives a bonus to narrow classes. Class 1 is five times narrower than class 2, so its density is five times taller at its peak, and that is enough to overcome the extra distance.

Scanning the winning class across the axis gives the decision regions

region	decision
$x < 0.1990$	class 3
$0.1990 < x < 0.3786$	class 2
$0.3786 < x < 0.4253$	class 1
$x > 0.4253$	class 3

Table 2: MAP decision regions for equal priors.

Class 1 owns only the narrow window $[0.3786, 0.4253]$, of width 0.047, because its small σ makes its density collapse quickly away from 0.4; and because class 3 is so broad it claims everything far from the other two means, on both sides.

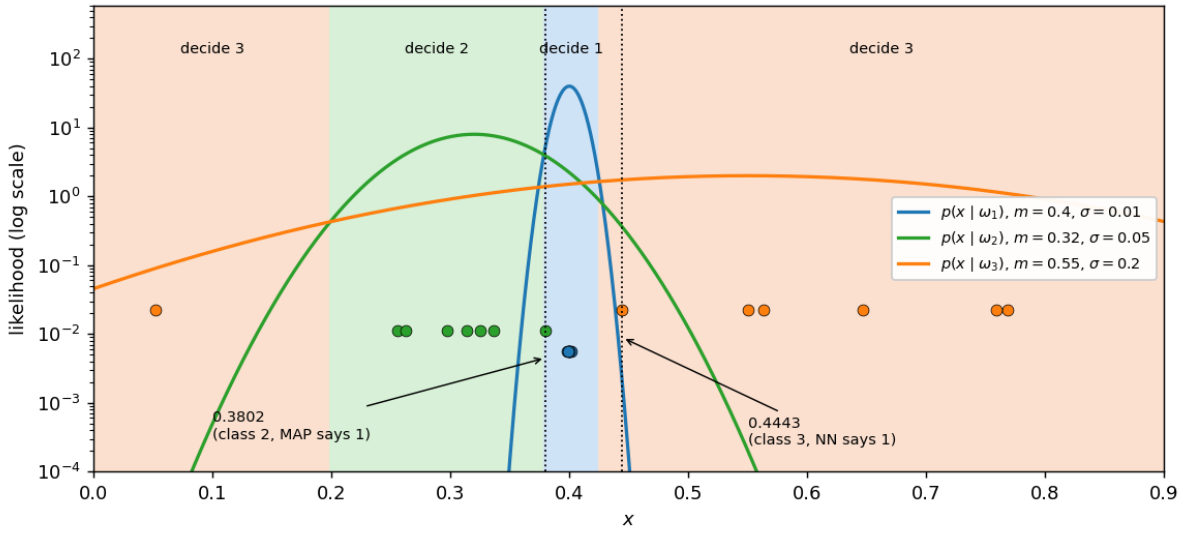


Figure 4: The three class likelihoods on a logarithmic scale, with the MAP decision regions shaded and the measurements shown as dots. The two misclassified values are marked.

Comparison of the two classifiers

Our predictions were right. Class 1 is classified perfectly by both methods, and the two errors fall in classes 2 and 3, the ones we expected to be harder. Each error has its own cause. Nearest neighbour misses 0.4443 because the four training samples of class 3 leave a wide gap between 0.05 and 0.56 and that point falls inside it. MAP misses 0.3802 because it is the most extreme sample of class 2 and reaches into the narrow window belonging to class 1.

The two methods can disagree about the same value because they measure closeness in different ways. Nearest neighbour uses the plain distance to the closest training sample. The Gaussian model divides that distance by the standard deviation of the class, so it asks how *surprising* the distance is rather than how large it is. For $x = 0.4443$ the plain distances are 0.0440 to class 1 and 0.1189 to class 3, so nearest neighbour picks class 1. Measured in standard deviations the same point is $4.43\sigma_1$ from m_1 but only $0.53\sigma_3$ from m_3, so MAP picks class 3 and is right. Nearest neighbour has no way of knowing that a deviation of 0.044 is ordinary for class 3 and impossible for class 1.

If the classes were not equally likely the priors would no longer cancel, and we would maximise $\ln f_X(x \mid Y = k) + \ln P(Y = k)$. Each class then gets a fixed bonus $\ln P(Y = k)$ that does not depend on x, so the curves keep their shape but every boundary moves towards the less likely class. A class with a small prior needs stronger evidence from the measurement before it is chosen.

4 Image Classification

A scene is one of three possible 2×2 binary images, with the a priori probabilities

$$Y = 1 : \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \ \tfrac{1}{4}, \qquad Y = 2 : \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \ \tfrac{1}{2}, \qquad Y = 3 : \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \ \tfrac{1}{4}.$$

One of them has been sent through a noisy channel that flips each pixel independently with probability ϵ, and the image we actually observe is

$$x = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Using Bayes theorem, calculate the posterior probabilities and the MAP estimate of the scene for (a) $\epsilon = 10\%$ and (b) $\epsilon = 50\%$. Show the answers in percentages or percentage decimals.

Since the noise acts on each pixel separately, the probability of seeing the whole image is the product of the probabilities of seeing each of its four pixels,

$$P(x \mid Y = k) = \prod_{j=1}^{4} P(x_j \mid Y = k, \text{pixel } j),$$

and each pixel contributes only one of two numbers: $1 - \epsilon$ if it agrees with the candidate scene, because then it came through unchanged, and ϵ if it disagrees, because then it must have been flipped. If the scene disagrees with the observed image in k of the four positions we therefore get

$$P(x \mid Y = k) = \epsilon^k \, (1 - \epsilon)^{4-k}.$$

Only the *number* of disagreements matters, not which pixels they are, because the flip probability is the same in both directions. Counting them by reading the images in the order top-left, top-right, bottom-left, bottom-right, where $x = (0, 1, 1, 1)$:

scene	pixels	prior $P(Y=k)$	mismatches k	likelihood
$Y=1$	$(0,1,1,0)$	$1/4$	1	$\epsilon(1-\epsilon)^3$
$Y=2$	$(1,0,0,1)$	$1/2$	3	$\epsilon^3(1-\epsilon)$
$Y=3$	$(1,1,1,0)$	$1/4$	2	$\epsilon^2(1-\epsilon)^2$

Bayes' theorem then gives

$$P(Y=k \mid x) = \frac{P(x \mid Y=k)\,P(Y=k)}{P(x)}, \qquad P(x) = \sum_j P(x \mid Y=j)\,P(Y=j),$$

so we compute the three products $P(x \mid Y=k)P(Y=k)$ and divide each by their sum.

(a) $\epsilon = 10\%$. The three products are

$$\tfrac{1}{4}(0.1)(0.9)^3 = 0.018225, \qquad \tfrac{1}{2}(0.1)^3(0.9) = 0.000450, \qquad \tfrac{1}{4}(0.1)^2(0.9)^2 = 0.002025,$$

which sum to $P(x) = 0.020700$. Dividing:

	$Y=1$	$Y=2$	$Y=3$
prior	25%	50%	25%
posterior	**88.04**%	2.17%	9.78%

Table 3: Posteriors for $\epsilon = 10\%$; exactly 81/92, 1/46 and 9/92.

The MAP estimate is scene 1. Notice that this overturns the prior: scene 2 was the most likely one before we looked, with 50%, but it drops to 2%, because for scene 2 to have produced this image three of the four pixels would have had to flip, and at $\epsilon = 10\%$ that is very unlikely. Scene 1 needs only one flip, so it rises from 25% to 88%.

(b) $\epsilon = 50\%$. Now $\epsilon = 1-\epsilon = 0.5$, so

$$P(x \mid Y=k) = (0.5)^k(0.5)^{4-k} = (0.5)^4 = \tfrac{1}{16}$$

for every scene: the number of mismatches cancels out. All three likelihoods are equal, so they are a common factor and drop out of Bayes' theorem, leaving

	$Y=1$	$Y=2$	$Y=3$
prior	25%	50%	25%
posterior	25%	**50**%	25%

Table 4: Posteriors for $\epsilon = 50\%$: identical to the priors.

The MAP estimate is scene 2, which is simply the scene that was most likely beforehand.

Comparing the two noise levels. What decides the balance is the ratio $\epsilon/(1-\epsilon)$, which we can call the price of one mismatch. If two scenes differ by one extra mismatch, their likelihoods differ by exactly this factor. At $\epsilon = 10\%$ the ratio is 1/9, so every extra mismatch makes a scene nine times less plausible and the observed image easily outweighs the priors. At $\epsilon = 50\%$ the ratio is 1, so mismatches cost nothing at all and the priors are left to decide on their own.

Between the two extremes the answer switches over: scenes 1 and 2 are equally good when $\frac{1}{4}\epsilon(1-\epsilon)^3 = \frac{1}{2}\epsilon^3(1-\epsilon)$, that is when $(1-\epsilon)^2 = 2\epsilon^2$, which gives

$$\epsilon = \frac{1}{1+\sqrt{2}} = \sqrt{2} - 1 \approx 0.414.$$

Below that noise level the evidence wins and we answer scene 1; above it the prior wins and we answer scene 2.

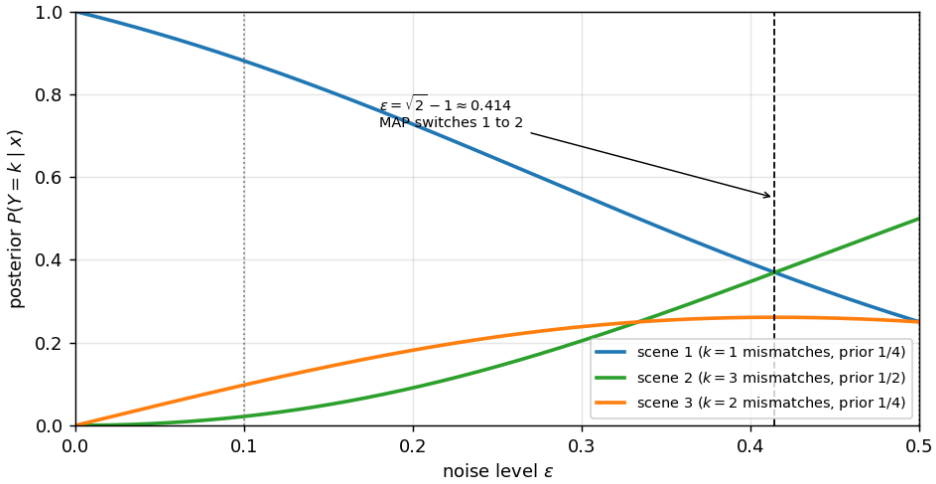


Figure 5: The three posteriors as functions of the noise level. At $\epsilon = 0$ the observation decides completely and scene 1 has probability 1; as ϵ grows the three curves move towards the priors 25%, 50%, 25%, which they reach at $\epsilon = 0.5$.

What $\epsilon = 50\%$ means. At this noise level every pixel is an independent coin toss, so any observed image is equally likely no matter which scene was sent. The observation therefore carries no information about the scene at all: looking at the image teaches us nothing, and the posterior is exactly what we believed before looking.

It is worth adding that $\epsilon = 50\%$ is the *only* noise level with this property. A noise level above 50% would be informative again, because then the pixels would be reliably wrong rather than random and we could recover the scene by inverting them. At $\epsilon = 90\%$, for example, the posterior of scene 2 is 94.2%, since a scene that disagrees with the observation in three of its four pixels is exactly what we expect to see when most pixels have been flipped.

5 Line Classification

The scene is a binary 4×4 image which is white everywhere except for one fully black column, and every pixel is observed wrongly with probability $\epsilon = 0.2$ in both directions. There are four possible scenes, one for each column, and their a priori probabilities are

$$P(Y = 1) = 0.3, \qquad P(Y = 2) = 0.2, \qquad P(Y = 3) = 0.2, \qquad P(Y = 4) = 0.3,$$

so the outer columns are more likely than the inner ones. In the observed image the black pixels sit at positions $(1, 1)$, $(2, 2)$, $(3, 3)$ and $(4, 2)$, writing $(\text{row}, \text{column})$.

Calculate the maximum a posteriori estimation of the given image when $\epsilon = 0.2$.

Before computing, it is worth guessing the answer from the picture alone. Column 2 contains two of the four observed black pixels, columns 1 and 3 contain one each, and column 4 contains

none, so we expect column 2 to be the best candidate, columns 1 and 3 to be clearly worse, and column 4 to be worst. We also expect columns 1 and 3 to be close to each other, since the pixel evidence treats them alike.

The likelihood is built exactly as in the previous section. The pixels are independent, and each one contributes $1-\epsilon$ if it matches the candidate scene and ϵ if it does not, so a candidate that disagrees with the observation in k of the sixteen pixels has

$$P(x \mid Y = c) = \epsilon^{k}\,(1-\epsilon)^{16-k}.$$

Counting the disagreements for each candidate gives $k = 6, 4, 6, 8$, as marked in Figure 6. Bayes' theorem then gives

$$P(Y = c \mid x) = \frac{P(x \mid Y = c)\,P(Y = c)}{P(x)}, \qquad P(x) = \sum_{j} P(x \mid Y = j)\,P(Y = j).$$

For column 2, for example,

$$P(x \mid Y = 2) = 0.2^{4} \cdot 0.8^{12} = 1.0995 \times 10^{-4},$$
$$P(x \mid Y = 2)\,P(Y = 2) = 0.2 \cdot 1.0995 \times 10^{-4} = 2.1990 \times 10^{-5}.$$

Doing the same for all four and dividing by their sum $P(x) = 2.5555 \times 10^{-5}$:

column c	prior $P(Y=c)$	mismatches k	$P(x \mid Y=c)P(Y=c)$	posterior
1	0.3	6	2.0616×10^{-6}	8.07%
2	0.2	4	2.1990×10^{-5}	**86.05%**
3	0.2	6	1.3744×10^{-6}	5.38%
4	0.3	8	1.2885×10^{-7}	0.50%

Table 5: Posterior probabilities of the four line positions for $\epsilon = 0.2$.

The MAP estimate is a line in **column 2**, and the ranking $2 > 1 > 3 > 4$ is exactly the one we predicted from the picture.

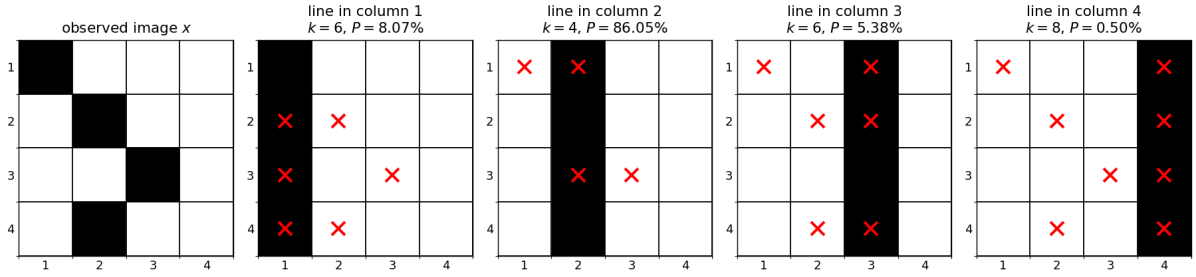


Figure 6: The observed image and the four candidate scenes. Red crosses mark the pixels where a candidate disagrees with the observation.

Evidence for and against two candidates. For column 2 the evidence in favour is the pair of black pixels at $(2,2)$ and $(4,2)$, which are exactly where the line should be. The evidence against is of two kinds: the pixels $(1,2)$ and $(3,2)$ are white although the line should make them black, and the pixels $(1,1)$ and $(3,3)$ are black although they lie outside the line and should be white. That is four disagreements in total, fewer than any other candidate.

For column 4 there is no evidence in favour at all, since none of the observed black pixels lies in that column. Against it, all four pixels of column 4 are white where the line should have made them black, and all four observed black pixels are outside the column and should have been white. This gives eight disagreements, which is the worst score of the four candidates and explains its posterior of only 0.50% despite its high prior.

The effect of the unequal priors. The priors and the pixel evidence point at different candidates here. The priors favour the outer columns 1 and 4 with 0.3 each, while the evidence favours column 2, which has the fewest disagreements. The evidence wins, and we can see by how much. Column 2 has two disagreements fewer than column 1, and since each disagreement costs a factor $\epsilon/(1-\epsilon) = 0.25$, the likelihood of column 2 is $4^2 = 16$ times larger. The prior works the other way but only by a factor $0.2/0.3 = 2/3$, so the posterior ratio is $16 \cdot \frac{2}{3} \approx 10.7$, which matches 86.05/8.07 from the table. Two extra matching pixels are simply worth far more here than the prior preference for an outer column.

The priors are not irrelevant, though, and there is one place where they decide everything. Columns 1 and 3 both disagree with the observation in six pixels, so their likelihoods are identical and the pixel evidence cannot separate them at all. What separates them is only the prior, and indeed their posteriors stand in the ratio $8.07/5.38 = 1.50$, which is exactly $0.3/0.2$. The prior acts as a tie-breaker when the image alone is not informative, but it is easily overruled when the image is.

6 Character Classification

An image of 5×3 pixels shows one of three characters, with a priori probabilities

$$P(Y = \text{‘B’}) = 0.30, \qquad P(Y = \text{‘0’}) = 0.45, \qquad P(Y = \text{‘8’}) = 0.25.$$

The noise is different from the previous two sections, because it is no longer symmetric: a truly white pixel is measured as black with probability 0.35, while a truly black pixel is measured as white with probability 0.2. Writing the four cases out,

$$\begin{aligned} &P(\text{white} \mid \text{truly white}) = 0.65, && P(\text{black} \mid \text{truly white}) = 0.35, \\ &P(\text{white} \mid \text{truly black}) = 0.20, && P(\text{black} \mid \text{truly black}) = 0.80. \end{aligned}$$

How should the image x be classified using maximum a posteriori probabilities (MAP) in the classification criterion?

As before the pixels are independent, so the likelihood of the whole image is the product of fifteen single-pixel probabilities, each read off the table above according to what the template says and what we observed. The important difference from Sections 4 and 5 is that we can no longer just count mismatches. There the two kinds of error had the same probability, so only the total number mattered; here a mismatch on a white template pixel contributes 0.35 while a mismatch on a black template pixel contributes 0.20, so the same number of mismatches can give quite different likelihoods depending on which kind they are. Mismatching a black pixel is $0.35/0.20 = 1.75$ times more damaging than mismatching a white one. We therefore sort the fifteen pixels into four groups and write

$$P(x \mid Y = k) = 0.65^a \cdot 0.35^b \cdot 0.20^c \cdot 0.80^d,$$

where a counts the pixels that are white in the template and white in x, b white in the template but black in x, c black in the template but white in x, and d black in both.

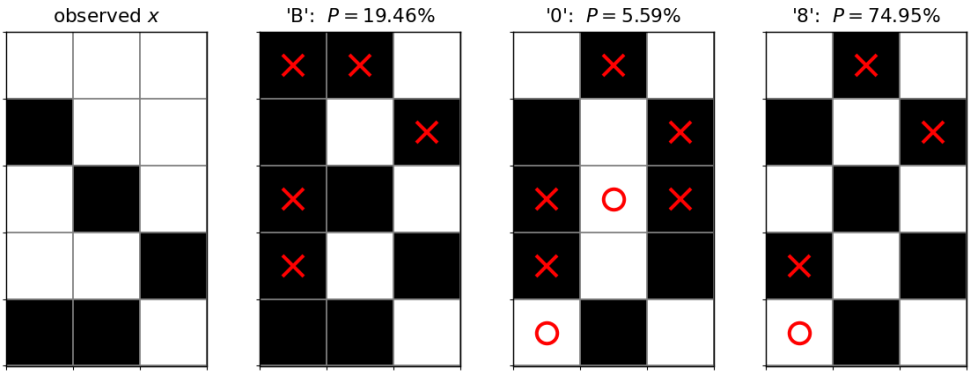


Figure 7: The observed image and the three templates. A cross marks a black template pixel observed as white (contributing 0.20), a circle marks a white template pixel observed as black (contributing 0.35).

Counting the four groups from Figure 7 and multiplying by the priors gives

	a (0.65)	b (0.35)	c (0.20)	d (0.80)	$P(x \mid Y = k)$	$P(x \mid Y = k)P(Y = k)$	posterior
'B'	5	0	5	5	1.2167×10^{-5}	3.6500×10^{-6}	19.46%
'0'	5	2	5	3	2.3288×10^{-6}	1.0479×10^{-6}	5.59%
'8'	7	1	3	4	5.6223×10^{-5}	1.4056×10^{-5}	**74.95%**

Table 6: Posterior probabilities of the three characters, with $P(x) = 1.8754 \times 10^{-5}$.

The MAP classification is **'8'**.

Where the evidence comes from. The observed image has five black pixels, while the templates have seven ('8'), eight ('0') and ten ('B'). Since a black template pixel appearing white is the expensive kind of error, a template with many black pixels has a lot to explain away, and this is what decides the result. The character 'B' actually fits the black pixels perfectly: every one of the five black pixels in x falls on a black pixel of 'B', so $b = 0$ and there is no evidence of the cheap kind against it. But 'B' also predicts ten black pixels and only five appeared, leaving $c = 5$ black pixels unaccounted for, and five factors of 0.20 is a heavy penalty. The character '8' has one pixel of the wrong kind, the black pixel at the bottom left of x which '8' expects to be white, but it only has to explain away three missing black pixels instead of five. Two fewer factors of 0.20 is worth a factor 25, far more than the single factor 0.35 costs, so '8' ends up about 4.6 times more likely than 'B'. The character '0' is worst of the three: it has both two pixels of the wrong kind and five missing black pixels.

Prior against evidence. The final classification is determined almost entirely by the observed pattern, not by the priors. The prior actually points the other way: '0' is the most likely character beforehand at 45%, yet it finishes last with 5.59%, while '8' was the least likely beforehand at 25% and finishes first with 74.95%. Comparing those two directly, the likelihood ratio is $5.6223/0.2329 \approx 24$ in favour of '8', whereas the prior ratio is only $0.25/0.45 \approx 0.56$ in favour of '0'. Multiplying the two gives about 13.4, which is exactly the ratio 74.95/5.59 of the posteriors. The image evidence is therefore roughly twenty-four times stronger than the prior preference is, and it comfortably overturns it.

7 The Garment Classification system – part 2 – Feature extraction

The task is to write `segment2feature`, which takes one binary segment and returns a feature vector. The dataset `short1` contains ten images of five garments each, so fifty segments in total, and the characters in the ground-truth files are garment class indices rather than digits: 0 T-shirt, 1 trouser, 2 pullover, 3 dress, 4 coat, 5 sandal, 6 shirt, 7 sneaker, 8 bag and 9 ankle boot.

> Given a binary image B, study the region of pixels that are equal to one. Define at least six different features that can be used to classify which garment class the region corresponds to. It is good if the feature does not depend on which position the region has. Describe and motivate the features you have chosen, stating what property of a garment each is intended to capture and what invariances you expect it to have.

We use ten features. Each one is computed either from the segment after cropping it to its bounding box, or from central moments, which are measured relative to the centre of mass. Both constructions ignore where the garment sits in the frame, so **every feature is translation invariant**: shifting a segment in x or y gives exactly the same vector. Features 2–4 are in addition scale invariant, since they are divided by $\sqrt{\text{area}}$, and features 1, 7, 9 and 10 are ratios and so are scale invariant as well. None of them is rotation invariant, which we return to below.

#	feature	what it is meant to capture
1	aspect	$h/(h+w)$ of the bounding box. Tall narrow garments such as trousers and dresses score high, footwear scores low.
2	extent	area divided by bounding-box area. A pullover fills its box, while trousers leave a notch between the legs and a sandal is mostly gaps.
3	spread_x	standard deviation of the pixels in x, divided by $\sqrt{\text{area}}$: a normalised second central moment.
4	spread_y	the same in y. Together with 3 this describes how elongated the shape is, independently of its size.
5	tilt	$\mu_{11}/\sqrt{\mu_{20}\mu_{02}}$, in $[-1, 1]$. Zero for a shape symmetric about a vertical axis; clearly negative for footwear, which lies diagonally in the frame.
6	holes	number of enclosed background regions, capped at two and divided by two. A bag with a handle encloses a hole.
7	compactness	$4\pi\,\text{area}/\text{perimeter}^2$, equal to one for a disc and small for a ragged boundary such as a sandal.
8	v_centre	height of the centre of mass inside the bounding box, 0 at the top and 1 at the bottom. Separates top-heavy from bottom-heavy garments.
9	width_25	width of the silhouette a quarter of the way down, relative to the widest row.
10	width_75	the same three quarters of the way down. Together these two sketch the outline: a shirt is wide at the shoulders, a dress widens towards the hem.

Table 7: The ten features. All are scaled to roughly $[0, 1]$, except the tilt which lies in $[-1, 1]$, so that no single feature dominates a distance computation.

Before running the benchmark, formulate at least two concrete expectations about the feature vectors: one concerning different images of the same garment class and one concerning images from different classes.

Expectation 1 (same class). Two shirts taken from different images should give similar vectors, and in particular should agree closely on aspect, v_centre and the two width features, since all shirts have roughly the same outline.

Expectation 2 (different classes). A trouser and a sandal should differ strongly in at least two features. We expect the trouser to have a much higher aspect, because it is tall and narrow while a sandal is wide and flat, and we expect the tilt to separate them too, since a sandal lies diagonally while a trouser stands upright.

Supply your code and representative examples of input segments with their corresponding feature vectors, including both different images of the same class and examples from different classes. Use these examples and the benchmark visualisation to assess your expectations. Identify features that appear discriminative and features that vary more than you expected within a class.

The code is in `assignment2.py`, and `garment_features.py` runs it over the whole dataset

and produces the figures below. Figure 8 shows two shirts from different images together with a trouser and a sandal, and their vectors are

	shirt A	shirt B	trouser	sandal
aspect	0.57	0.53	0.72	0.30
extent	0.84	0.56	0.81	0.61
spread_x	0.25	0.27	0.18	0.51
spread_y	0.32	0.35	0.48	0.22
tilt	0.01	0.02	−0.02	−0.27
holes	0.00	0.00	0.00	0.50
compactness	0.97	0.47	0.50	0.37
v_centre	0.54	0.54	0.44	0.55
width_25	0.90	0.91	1.00	0.59
width_75	1.00	0.65	0.82	1.00

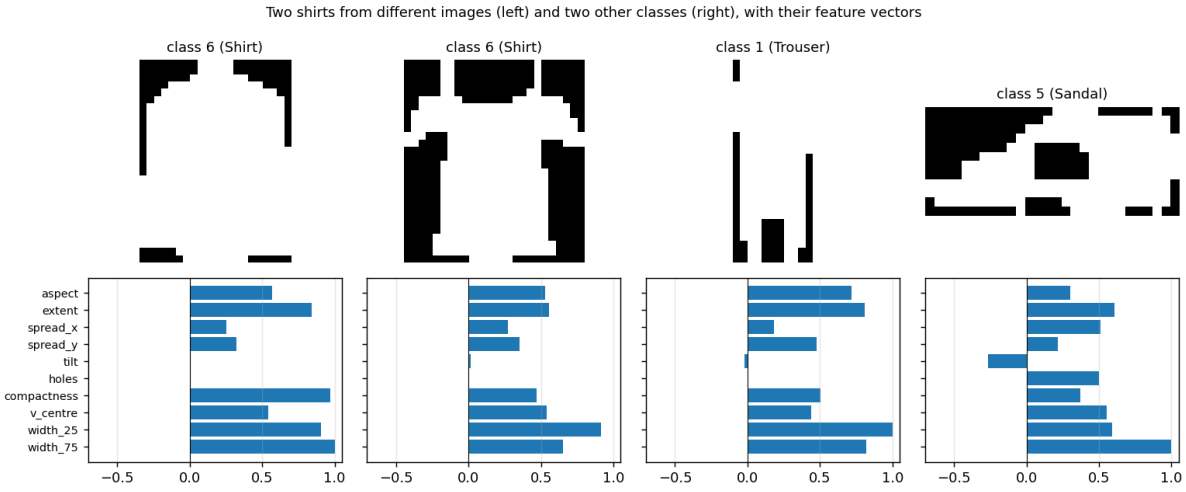


Figure 8: Two shirts from different images and two other classes, with their feature vectors.

Both expectations are largely confirmed. The two shirts agree to within 0.04 on aspect, exactly on v_centre and to within 0.01 on width_25, as predicted. The trouser and the sandal differ by 0.42 in aspect and by 0.25 in tilt, which is a much larger gap than anything we see within a class. But the shirts do *not* agree on everything: their compactness is 0.97 against 0.47 and their extent 0.84 against 0.56, which is a far bigger within-class difference than we expected. The reason is visible in the figure: in shirt B the sleeves stand away from the body, so the silhouette has deep notches under the arms, while in shirt A they do not. Both the fill fraction and the length of the boundary change a lot as a result.

To make this precise we compared, for each feature, the average spread within a class with the spread between class means. The ratio of the two says how useful a feature is:

holes	width_25	spread_x	aspect	tilt	spread_y	v_centre	extent	compact.	width_75
6.5	4.3	4.1	4.0	3.4	2.8	2.5	2.1	1.6	1.1

Table 8: Between-class spread divided by within-class spread, largest first.

The most discriminative features are the number of holes, which by itself identifies bags, followed by width_25, spread_x, aspect and tilt. At the other end, width_75 has a ratio of only

1.1, meaning it varies almost as much inside a class as between classes, and compactness is not much better at 1.6. These are exactly the two features that disagreed between the two shirts.

Running leave-one-out nearest neighbour on the standardised features classifies **29 of the 50 segments correctly**, or 58%, against 10% for guessing. Trousers and bags are recognised perfectly, 5/5 and 3/3, while pullovers are never recognised, 0/3.

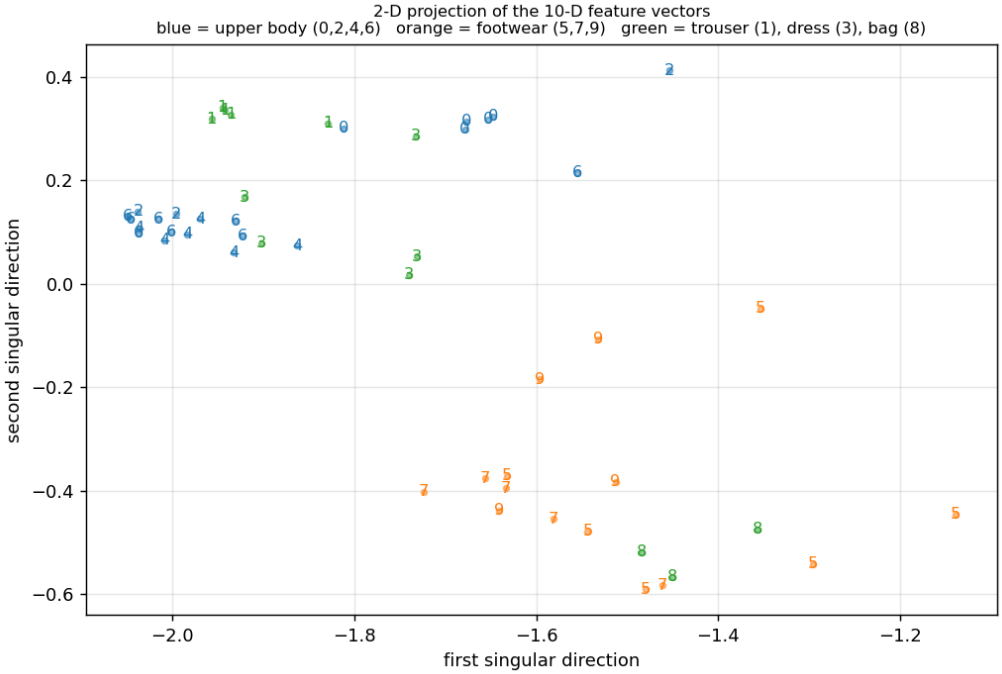


Figure 9: The benchmark's two-dimensional projection of the feature vectors. The three families separate clearly, but the classes inside each family overlap.

The projection in Figure 9 shows what is happening. The upper-body garments form one tight cluster on the left, the footwear forms another at the bottom, and trousers, dresses and bags sit apart from both. The *families* are well separated, but inside each family the classes are piled on top of one another.

Discuss at least one plausible failure mode: give an example of two garment shapes that could obtain similar feature vectors, or one garment whose feature vector could change substantially because of shape, scale, rotation, segmentation error, or another relevant variation.

The clearest failure mode is the pair coat and shirt. Both are upper-body garments with two sleeves and a straight hem, so their silhouettes are almost the same, and every one of our features is computed from the silhouette alone. The benchmark confirms this: three coats are classified as shirts and two shirts as coats, the two largest confusions in the whole dataset. Taken as a group, the four upper-body classes get only 11 of 22 right, and nine of the eleven errors are to another class *within* the group. Footwear behaves the same way: 7 of 15 correct, and all eight errors stay inside the footwear group. What actually distinguishes a coat from a shirt is the front opening, the buttons and the texture of the fabric, none of which survives in a binary mask, so this failure is a limitation of the input rather than of our particular choice of features.

A second failure mode concerns rotation. None of our features is rotation invariant, and two of them, tilt and aspect, would change a great deal if a garment were laid down at a different angle. A sneaker photographed pointing the other way would flip the sign of its tilt from -0.27 to $+0.27$ and would then look, to the classifier, unlike any sneaker it has seen. The same applies

to segmentation errors: if the threshold from Assignment 1 clipped a sleeve, the bounding box would shrink and aspect, extent and both width features would all shift at once.

> If you were allowed to add or replace one feature after inspecting the results, explain what you would change and what weakness in the current representation you would try to address.

We would replace width_75. It is the weakest feature we have, with a between-class to within-class ratio of only 1.1, so it carries almost no information: nearly every garment is close to its widest at three quarters of its height, whether it is a coat, a shirt or a sneaker.

In its place we would use the full row-width profile, resampled to a fixed number of steps down the height and divided by the widest row. This is still translation and scale invariant, but instead of one number at one height it describes the whole outline, and it is the kind of detail that could separate a coat from a shirt: a coat has roughly straight sides, whereas a shirt tapers from the shoulders to the waist. Our current pair of width measurements samples that profile at only two points and misses the shape in between. This addresses the weakness that our features describe the silhouette only very coarsely, which is what makes all four upper-body classes collapse into a single cluster in Figure 9.